

WORKSHEET - NAPLAN Year 9

Number and Algebra Basic Calculations and Algebra (9)

Teacher: Joanne Rust

Exam Equivalent Time: 8.75 minutes (based on NAPLAN allocation of ~1.25 min/mark)



Questions

1. ☒ Number and Algebra, NAP-13229

Which expression is equal to $4a^4$?

- ☐ $8 \times a$
☐ $16 \times a$
☐ $4 \times a \times a \times a \times a$
☐ $4 + a + a + a + a$
☐ $4 \times 4 \times 4 \times 4 \times a \times a \times a \times a$

2. ☒ Number and Algebra, NAP-07315

Which is the correct expression for 96.4851 rounded to 2 decimal places?

- 96.00** **96.48** **96.49** **96.50**
☐ ☐ ☐ ☐

3. ☒ Number and Algebra, NAP-30835

Ali joins a gym that has a fortnightly fee of \$33.

Members pay an extra \$12 per session if they want to do a special yoga class.

Which expression represents Ali's fortnightly gym bill if she does x yoga classes during the fortnight.

- $12x$** **$33x$** **$33x + 12$** **$33 + 12x$**
☐ ☐ ☐ ☐

4. ☒ Number and Algebra, NAP-25118

When $p = 3$ and $q = -3$, what is the value of $p^2 + q^2$?

$p^2 + q^2 =$

5. ☒ Number and Algebra, NAP-19203

A positive number x is multiplied by 3, then 15 is added.

Which of the following operations give the same result?

- ☐ Adding 15 to x , then multiply by 3.
☐ Adding 3 to x , then multiply by 15.
☐ Adding 3 to x , then multiply by 5.
☐ Adding 5 to x , then multiply by 3.

6. ☒ Number and Algebra, NAP-66243

Which of the following is always equal to $a - b$?

- $-a - (-b)$ $-a + b$ $b - a$ $-b + a$
☐ ☐ ☐ ☐

7. ☒ Number and Algebra, NAP-83825

$x^2 > 5x$

What is the smallest positive whole number x can be that makes this expression correct?

Worked Solutions

1. Number and Algebra, NAP-13229

$$4a^4 = 4 \times a \times a \times a \times a$$

2. Number and Algebra, NAP-07315

$$96.49$$

3. Number and Algebra, NAP-30835

$$\begin{aligned}\text{Fortnightly cost} &= \text{fee} + \text{yoga classes} \\ &= 33 + 12x\end{aligned}$$

4. Number and Algebra, NAP-25118

$$\begin{aligned}p^2 + q^2 &= 3^2 + (-3)^2 \\ &= 9 + 9 \\ &= 18\end{aligned}$$

5. Number and Algebra, NAP-19203

$$\text{Number} = 3x + 15$$

Consider each option:

$$(15 + x) \times 3 = 3x + 45$$

$$(3 + x) \times 15 = 15x + 45$$

$$(3 + x) \times 5 = 5x + 15$$

$$(5 + x) \times 3 = 3x + 15 \quad \checkmark$$

6. Number and Algebra, NAP-66243

$$a - b = -b + a$$

Worked Solutions

7. Number and Algebra, NAP-83825

$$\text{If } x = 6,$$

$$6^2 > 5 \times 6$$

$$36 > 30 \Rightarrow (\text{correct})$$

All positive whole numbers smaller than 6 result in:

$$x^2 = 5x \quad (\text{for } x = 5)$$

$$x^2 < 5x \quad (\text{for } x = 4, 3, 2, \dots)$$